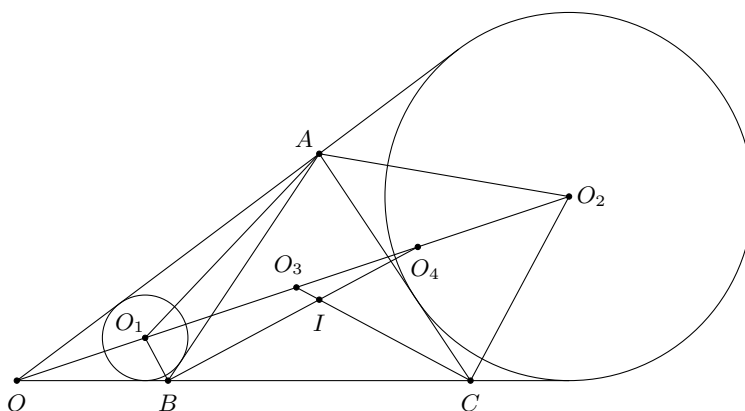


Problem 25. Point A lies on one ray and points B, C lie on the other ray of an angle with the vertex at O such that B lies between O and C . Let O_1 be the incentre of $\triangle OAB$ and O_2 be the center of the excircle of $\triangle OAC$ touching side AC . Prove that if $O_1A = O_2A$, then the triangle ABC is isosceles.

Solution.



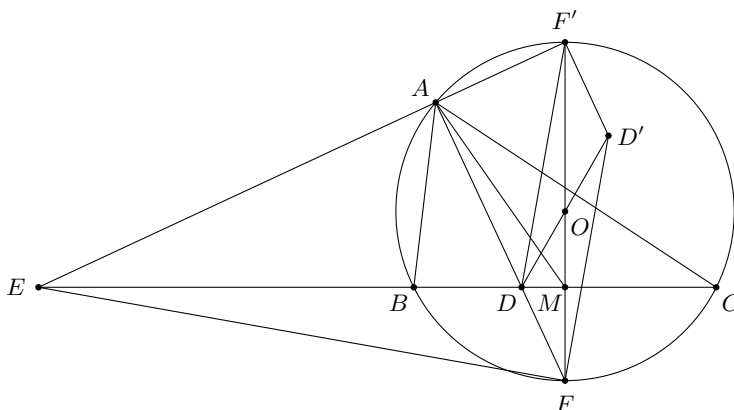
Let O_3 be the incentre of $\triangle OAC$, O_4 the excentre of $\triangle OAB$ opposite to O , and I the incentre of $\triangle ABC$. Note that I is the intersection of BO_4 and CO_3 , and O, O_1, O_2, O_3, O_4 are collinear.

As O_1 and O_4 are the incentre and the O -excentre of $\triangle OAB$, the points A, O_1, B, O_4 are concyclic. This implies $\angle O_4BA = \angle O_4O_1A$. Similarly, as O_3 and O_2 are the incentre and the O -excentre of $\triangle OAC$, the points A, O_3, C, O_2 are concyclic. Therefore, we find that $\angle ACO_3 = \angle AO_2O_3$.

If $O_1A = O_2A$, then $\angle AO_2O_1 = \angle O_2O_1A$. Thus, $\angle ACI = \angle AO_2O_1 = \angle O_2O_1A = \angle IBA$. This implies $\angle ACB = \angle CBA$ so that $\triangle ABC$ is isosceles. □

Problem 26. In $\triangle ABC$, the interior and exterior angle bisectors of $\angle BAC$ intersect the line BC at D and E , respectively. Let F be the second point of intersection of the line AD with the circumcircle of $\triangle ABC$. Let O be the circumcentre of $\triangle ABC$ and let D' be the reflection of D in O . Prove that $\angle D'FE = 90^\circ$.

Solution 1.



Note that $AB \neq AC$, since otherwise the exterior angle bisector of $\angle BAC$ would be parallel to BC . So assume without loss of generality that $AB < AC$. Let M be the midpoint of BC and let F' the reflection of F in O , which is also the second intersection of the line AE with the circumcircle of $\triangle ABC$. We now have

$$\angle D'FO = \angle OF'D.$$

